

Pareto-Nash Signalling with Nonlinear Peer Effects: Analysing the Microeconomics of Inclusion

MSc in HRM (Open) Dissertation

Supervisor: Dr. Nicola J. Bowman

Submitted to The Open University Business School in partial fulfilment of Master of Science in HRM
(CIPD Accredited)

Declaration

I declare that this dissertation was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text. This work has not been presented for any other degree or professional certification.

Signed:  _____

Dated: **20.03.2026**

Thesis Title (2nd Draft):**"Pareto-Nash Signalling with Nonlinear Peer Effects: Analysing the Microeconomics of Inclusion"**

Alfred. A. B. Mayaki¹

Author Correspondence

Open University Business School
The Open University
Milton Keynes, MK7 6AA
Buckinghamshire, England
United Kingdom**Abstract**

By assuming the view that signalling practices are organizational cues designed to communicate important strategic initiatives to workers, this Master's thesis analyses N=1,034 survey responses using a *two*-staged least squared (2SLS) regression framework and instrumental variable (IV) approach. The thesis presents a study into the stability of a Pareto-Nash equilibrium between firm outlay on wages under optimality conditions and a 'many-to-one' signal using data sampled on observed wages per productive worker.

Keywords: Information design, signalling, corporate communication, 2SLS, quantitative methods, econometrics, instrumental variable (IV)

JEL: J24, J71, D82, D83, D86, D90, M51, M54, I12, I14

Introduction

Artificial Intelligence (Henceforth "AI") has been argued to carry the potential to change or affect various types of skill-based worker allocations, to mitigate the consequence of overexposure to unconscious biases, and to help design more comprehensive approaches to allocate talent, and efficiently categorise new workers to roles, whilst developing an overarching organisational focus on *skill-based* outcomes, more generally. However, according to Kolotilin, *et. al.* (2025), a peer effects problem exists, particularly with respect to the task of allocation under Bayesian persuasion (Gentzkow and Kamenica, 2011; Galeotti *et.al.*, 2025). One of the more pronounced gaps in the literature on AI is competition and negotiated transfers (Mayaki, 2024; Smith *et. al.*, 2025) which doesn't correct for the capture of nonlinear equilibria.

Theoretical Background

While AI and signalling are two very similar but distinct topics, a great starting point for critique is the work of Kolotilin *et. al.* (2025) who defines two explicit cases of the 'Kamenica-Gentzkow' or 'KG' Bayesian persuasion model (Gentzkow and Kamenica, 2011; Che *et. al.*, 2022) in his work—Bayesian persuasion is defined by Kolotilin as a situation where many senders attempt to persuade a single receiver to take a particular action (in this case, to hire, given a revealed signal about the sender's circumstances)—one case being the linear, and the other being the *nonlinear* occurring in this respect, with a random walk² variable (Skardal and Adhikari, 2019). This occurs with an assumption of no misinformation. In a written work on signalling and its numerous contexts, another cohort of authors find evidence—

¹All author correspondence regarding this thesis should be made to the following e-mail address: alfred.mayaki@ou.ac.uk. The author is a Royal Economic Society student member.

² Notwithstanding the case of the *linear* random walk where the variable assumes independent and identically distributed properties

in a review of digitalisation—that a *linear* process (Gentzkow and Kamenica, 2016) has several shortcomings when viewed in its *traditional* sense. If we assume the linear case i.e. a *frictionless* signalling environment or a case where the sender and receiver are able to successfully design a signal message that maximises the probability of signal realization via the message, then the premise of this study is rather trivial, and the question is answered quite formally in the proofs in Kolotilin *et. al.*'s outstanding commentary. However, if we consider the *nonlinear* case with costly frictions (Che, Kim and Mierendorff, 2022) for both the sender and receiver, then the problem of peer effects begins to arise.

Optimal Signalling and Strategic Form

Without an evidence-base and given the nuances of the Stages model, we cannot rely on an argument that it may be that the nonlinear case welcomes more readily strategic narratives or unitarian forms of stakeholder capitalism especially under the resource-based view (Lado and Wilson, 1994), where the owner incentives and competitiveness are prioritised but as Chapter 8 of Van Riel and Fombrun (2007) indicates, shareholders are not the only stakeholders in the organization with incentive to act in its best interests (what Lado and Wilson refer to as 'proclivity'), workers too, can be viewed as *internal* stakeholders. Signalling theory (Spence, 1973) therefore, widely seen as a seminal work (van Damme and Güth, 1991, p.264; Kolotilin, 2015), postulates itself as providing a microscopic lens into the origins of worker-firm incentive matching in economic theory, demonstrating how *unobserved* productivity between workers and firms (Bafera and Kleinert, 2023) sets the scene for a cogent analysis of competing interests with respect to a many-to-one job market solution (Kolotilin *et. al.*, 2024; Illyushina, Walker and Timming, 2024) where the peer effects problem persists. It remains the case that Spence (1973)'s main hypothesis—that signals act as communicative devices between firm and worker, such as when workers invest *more* in education signalling to firms that a higher wage is expected (Kolotilin, 2015)—is yet to be reconciled directly with the problem of peer effects and recent work on optimal worker allocation. The central challenge for this thesis involves interpreting an N=1,034 participant survey, presenting survey³ results using

descriptive statistics and summarising research which contextualises signalling theory in favour of the firm (Bafera and Kleinert, 2023) and in the context of the worker's interests (Miller and Doyle, 2025) at equilibrium. How workers observe inclusive strategies both internally and externally in a volatile and uncertain environment (Barney, 1991) links back to the art of optimal signalling throughout the employee '*journey*' or the workforce lifecycle (Celani and Singh, 2011; Miller and Doyle, 2025). The study investigates several reoccurring themes in the work of Celani and Singh (2011), and Miller and Doyle (2025) with the premise of a research article on signalling theory forming a large part of the literature review. The thesis finally undertakes a 2SLS regression analysis (Greene, 2003; Chen, 2019), with an instrumental variable (IV).

In emerging economies such as West Africa, with the rise of AI, the central planner's problem changes, from shifting the unemployment rate down to increasing the level of labor force participation. Let us firstly assume a low participation rate in an open economy model and that a hypothetical multinational firm F_N at time (t) , seeks to employ a cohort of n_M line managers, and n_N non-supervisory staff members at a new geographical location, with a given allocated cost function which we may refer to as a wage allowance or ex-ante wage budget $[B_N(t) \cdot (w)_N(t)]$. The shareholders of the firm, denoted as $S_F(t)$, vote unanimously by motion to hire a cohort of line managers and non-supervisory workers, closely managed by a team of expatriated and embedded human resource managers n_H in order to facilitate a geographical expansion of the firm into a new geographical location. New task roles and role profiles for all three categories of work (line managerial workers, non-supervisory staff, and human resource professionals) are identified, and a headcount figure is estimated for each role as part of a workforce plan.

Utility, Practices and Payoffs

The key feature of this model is each worker in the firm has an average tenure of γ years in employment. As in Mayaki (2025) initial firm endowments are exogenously determined. The open economy is therefore assumed to be perfectly competitive. The firm's optimisation problem is

³The participant survey's core questions are available to view online: <https://alfredmayaki.me/msc-survey>

initially a labor market allocation problem: that is, given an *estimated* annual turnover of $R_F(t)$, what should the optimal ex-ante wage budget $[B_N(t) \cdot (w)_N(t)]$ be to ensure all three cohorts of workers are sufficiently qualified, productive, and that the firm earns a reasonable percentage of competitive market share in its first year of trading as a market entrant? We consider a case where there are working practices that dictate the value of workers within the supply chain. The solution to this labor market problem of allocation of workers is of course partly dependent on the size of the firm's wage budget $B_N(t) \cdot (w)_N(t)$, which is a function of the firm's estimated annual turnover $R_F(t)$. However, as one might argue, forms of strategic management, narrative⁴ and choice rather than simply the offered wage is at least a partial consideration for incoming workers, company shareholders and senior management alike (Porter, 2008; Anderson and Markides, 2007).

The Nonlinear Case - Extending Kolotilin *et. al.* on Peer Effects

How then is labor force participation treated as a granular datapoint by firm $F_N(t)$? Prescriptively or normatively? Let us assume the workforce is allocated in a *nonlinear* way to a firm $F_N(t)$ on the basis of productive capacity (Kolotilin *et. al.*, 2024) the cumulative distribution of which is strictly quasi-concave, non-decreasing, continuous and homogenous of degree 1 in w_N at time t (Lambotte, 2025: p. 620; Varian, 1992: p.72). We can assume also that there are a finite number of workers, each with preference utility $U_N(t)$ and an infinite number of allocations $\sigma_F(t)$ unique to each firm in an economy and a finite number of priced transfers, which occur at a realised cost to each firm. Working practices, which we denote as $P_N(t)$, which are either skill or competency-based and finally many-to-one signalling occurs in regard to worker opportunity sets. Table 1.1 demonstrates the Pure Nash strategy for a simple two player game of priced transfers where in the first stage co-workers signal intentions to commit to a level of effort and firms choose whether or not to hire the cohort.

Table 1.1: Nash Strategy Payoff Matrix

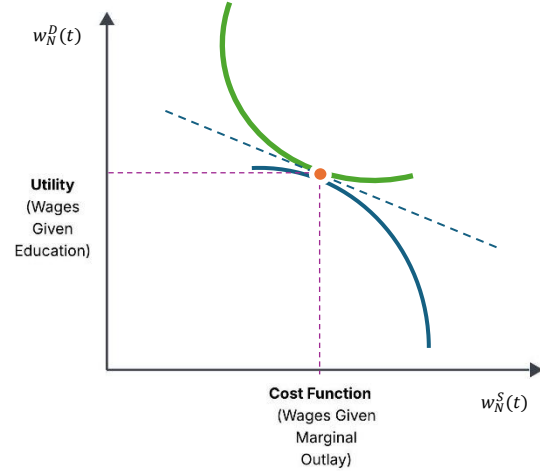
Table 1.1: Nash Strategy Payoff Matrix

<i>Workers and Firms</i>	Low Wage Offer (t)	High Wage Offer (t)
Low Utility (t)	w_N^{DL}, w_N^{SL}	w_N^{DL}, w_N^{SH}
High Utility (t)	w_N^{DH}, w_N^{SL}	w_N^{D*}, w_N^{S*}

Pre-Defining Equilibrium Conditions

We only assume competency-based working practices display monotonicity along the firm *output* function, prioritising worker education, proficiency and productive efficiency. All three worker types may only signal to firms their level of experience and education which signal productive capacity and a categorical wage utility. We can therefore produce an objective function, where the real wage supplied to N workers $w_N^S(t)$ at time t is equal to the total firm revenue $R_F(t)$ minus the firm specific ex-ante wage budget divided by the total number of workers and divided by a firm specific *scalar* of working practices, which is strictly concave. Hence, equilibrium conditions are met everywhere where:

Figure 1.1: Pure Nash Equilibrium



$$(1.11) \quad (\operatorname{argmin})w_N^{S*}(t) = (\operatorname{argmax})w_N^{D*}(t)$$

The wage budgeted by each firm is equal to the wage utility of N workers which is slightly easier to deduce, as the function $w_N^D(t)$ is composed, as such, of *exogenously* determined concepts and variables such as preference for

⁴ We use this term to generally imply the existence of any type of strategic planning within practice.

leisure and the ratio of household consumption-to-savings (Mayaki, 2024). The endogeneity in this thesis's model relates to the worker's pre-employment signal about their level of education E_U and the worker's pre-employment signal about their skills proficiency (i.e. level of experience) E_X multiplied by the likelihood that the worker is a member of a politically-affiliated trade union which is a percentage, $\mu_N(t)$, ranging between 0% (low likelihood of union affiliation) and 100% (certainty of union affiliation). Further empirical research may wish to explore the influence of peer effects on trade union membership under different equilibrium conditions.

Further Equilibrium Conditions

In a repeated game of optimal information design, we are particularly interested in solutions where $(\text{argmin})w_N^S(t) \neq (\text{argmax})w_N^D(t)$ occurs or rather where heterogeneous worker allocations (which are ordinal and ranked in group form based on education and productivity) are not in equilibrium as per Blandin, Boyd and Prescott (2015) and as indicated by the outcome of the negotiated wage and the worker's output per increment of time ($t \dots T$), where worker heterogeneity interjects with supermodularity and monotonicity of practices-to-transfers (Gentzkow and Kamenica, 2016; Kolotilin *et al.*, 2025), hence the sweet spot occurs where the boundary condition of strategic complementarity (Galeotti *et al.*, 2009: p.220) is always met (Figure 1.1) by all three conditions in equilibrium.

In an earlier work (Mayaki, 2024), I argue for two concepts; the first being the existence of a pre-game agreement between each firm and their workforce, allowing workers the privilege of discussing 'the *possibility* of entering into a contract for employment', whereby players may also create incentivised network links with co-workers as with Lambotte (2025) prior to engaging in the signalling game (Blume and Ortmann, 2007: p.279), with the incentive of improving transfer capacity measured by a shift in the preference utility for $w_N^D(t)$ where $w_N^D(t) > \sum_1^N W_N^S(t)$ occurs as the difference between signalled productivity and improvements in actual group productivity. The second being an equilibrium *across* markets, where firms may 'signal entry' and 'signal to workers' about the nature of its respective competitive strategy.

Relevant Literature Review

Identifying Gaps in the Research Literature

This Master's thesis focuses on the Pareto-sufficient nonlinear context of signalling in labor markets, though there are other concurrent contexts pertaining to the term signalling in the literature that reference similar works such as studies on language evolution (Scott-Phillips, Kirby and Ritchie, 2009), environmental pollution, with particular application to the '*shallow lake problem*' (Wagener, 2013) or studies on street gang recruitment (Densley, 2012) which describes 'strategies of obligation, coercion, seduction and subterfuge' used to 'attract new members'. The chief ambition here is not to mirror these alternative use cases, but rather to generalise concerning a stable equilibrium condition for the negotiated wage given signals about worker productivity in a scenario where workers may be pooled based on skills proficiency and education.

Here, Feld and Zölitz (2017) have found evidence in favour of peer effects in education. This is significant because considering education as a signal or at least a proxy for productivity is a valid assumption. The gap in the signalling literature on education is not very vast; however, it extends to recent work by Kolotilin *et al.* (2024) and with formal proofs with application to education, labor and media censorship. There is little or no example, however, before Kolotilin *et al.* (2024) of any recent empirical results showing survey evidence achieved through *primary* quantitative research. Comparing Kolotilin *et al.* (2024) to van Damme and Güth (1989), the work of the latter and thereafter van Damme and Güth (1991) explicitly refers to Spence (1973) in a model focused on perfect equilibria—as previously demonstrated in Mayaki (2024) via the notion of 'disagreement points' in what van Damme and Güth (1989) refer to as '*uniformly* perfect equilibria'—both worker and firm frictions are seen as a significant element to equilibrium search and matching models. Here, one might consider signalling as a solution concept to these frictions but likewise it may be feasible to offset the pros of signalling by arguing that equilibrium analysis of corporate communication and information design is too hypercritical, interested in detail which does not exist and is along the lines of the basic-hard approach, which fails to acknowledge basic tenets of human designed interactions (Van Riel and Fombrun, 2007).

Critiquing Spence's Signalling Game

In the view of Spence (1973) the notion of worker allocation is made analogous to the rationale of purchasing a ticket for a lottery draw, which itself becomes a problematic notion due to the euphemisms of risk, value and eventual loss being invoked. This rationale for adopting such a gamble is best described per a model composed by Barmby *et al.* (1994). Spence argues that there is often an element of informational asymmetry pertaining to full knowledge of the average workers' skill or proficiency prior to such a lottery concept being invoked. Assuming the lottery concept is associated with information that is rarely fully disseminated, a firm's perception of this lottery gamble is what Spence (1973) credits as his "primary interest" due to the uncertainty involved when weighed against the wages offered by the firm to the worker as compensation for services. When discussing the lottery, we speak here of a singular instance of a repeated game theoretic timeline. The terms of reference adhered to by Spence (1973), constitute very strict and formal 'belief systems' for his term sheet mapping the firm's observation of the lottery process. There are two kinds of attributes mentioned in Spence's article: 'indices' and 'signals' (Spence, 1973; p. 357). The term 'indices' refers to immutable attributes observed by the firm amongst participants in the hiring 'lottery', attributes such as race or sex. This is followed closely by the familiar term 'signals', which references alterable attributes or those which can be manipulated by the worker (to improve on), such as education and experience. For the purposes of Spence's research, age is considered as an alterable attribute and is therefore seen to be an index. Signals and indices are seen to be 'parameters' in the conditional probability distribution reflecting the 'beliefs' of a firm. Upon reflection with some experience, it's fair and proper to infer proportionality. By this, I mean the Spencian idea that work is a lottery, is not an appropriate microscopic lens to view employment dynamics when compared to the objective of improving education and labour force participation.

Synthesising Miller and Doyle and Spence's Signalling Game

Much of the recommendations of this thesis on the topic of optimising an inclusive workforce can be supported by academic literature. We identify the scenario and problem faced by adult workers. Miller and Doyle (2025) shed further light on this problematisation of a Spencian-styled 'lottery'

scenario taken by firms, providing what one might call a *synthesis* to the concerns highlighted in their paper. According to Miller and Doyle (2025) the article revisits the empirical findings of Khan *et al.* (2023) whose literature review and wider study on the topic is the only post-COVID era work used as authority, but the article also acknowledges that research in the area of inclusive signalling to workers is relatively limited. Miller and Doyle (2025) apply a systems-based analysis of 'signalling' theory (or rather a systematic approach based on academic literature) in the allocation process to uncover a view that describes the inclusion of the neurodiverse worker in the workplace. While this thesis supports this analysis, it also draws attention to the resource constraints inherent to highly competitive companies (Lado and Wilson, 1994). Millar and Doyle (2025) nevertheless argue organisations should go *beyond* the business case for an heterogeneous allocation of neurotypical worker and neurodiverse co-worker by making the signalling process itself clearer (Van Damme and Güth, 1989).

Critical Evaluation of Celani and Singh on Outcomes

One can divide the approach for achieving such an eventuality as follows. Firstly, the work of Miller and Doyle treats an interesting argument in favour of neuroinclusive approaches (Khan *et al.*, 2023) by encouraging proactivity from the firm's managers. If a firm provides a list of 'frequently requested adjustments', then this firm action is a proactive signal to workers who may require such adjustments to participate in the many-to-one pre-allocation stage. Likewise, by sufficiently educating recruiters and hiring managers involved in the act of recruitment and selection on what "appropriate *behaviours*" are, and by thereafter implementing a follow-up action: prioritising diversity, equity and inclusion over the Spencian view of hiring as a lottery, we arrive at a mechanism which appreciates inclusion as a priority. Critiques may argue this may create a consequentialist dependence on prior training investment. In contrast to Miller and Doyle (2025)'s unrelenting pragmatism on the subject of signalling best practice, Celani and Singh (2011) have conducted an analysis of signalling theory from a *multi-level* perspective (with the use of 'contextual effects'). In multi-level critique, the assessment of how signals are seen to reveal unknown characteristics of the organization become *more* stratified.

The theory posited provides links between individual and organisation-level signals from firm actions. Links are made between market signals and instrumental or symbolic worker inferences. Unlike Spence (1973), Celani and Singh (2011) are concerned with outcomes. They therefore take up a consequentialist position (a position which derives itself from the moral philosophy of consequentialism) – with a focus on outcomes. By introducing their topic in such a way, Celani and Singh (2011) conclude by challenging readers to study a gap in the signalling literature—signalling from a *recruiter* perspective i.e. from the *organisation's* perspective. This means to imply that the organisation does indeed perceive and internalise what Spence (1973) defines as ‘alterable’. Signalling theory when viewed through the Spencian microscopic lens is therefore arguably either angst or incongruent (Smith *et. al.*, 2025) on the one hand, or catalyst and predicator on the other. By contrasting the approaches reviewed in Celani and Singh (2011) with that of Miller and Doyle (2025) it is possible to observe how in hindsight both authors take completely opposing perspectives on signalling in the context of inclusion. Chang and Chin (2018) offer a third way – idiosyncrasy. They define signalling as the communication of unique information, or rather “the sending out of information that competitors cannot easily imitate” analysing signalling on firm websites over a three-year period. Signalling therefore becomes important in job markets where numerous firms compete to “*increase market share*” as Chang and Chin (2018) infer stressing the importance of firm actions in a separating equilibrium.

The workforce lifecycle, now known as the employee ‘*journey*’, under competence-based frameworks (Lado and Wilson, 1994) covers a vast amount of terrain including interviews, performance management and compensation, which are unique to every firm yet even so, an important contemporary aspect with respect to signalling theory is that it increasingly come under the remit of *Large Language Models* (LLMs)—where the problem of peer effects, which is almost always inherent to generative AI, continues to surface (Galdin and Silbert, 2025; p. 36). Because competency-based frameworks are about optimising skills proficiency within the firm, signalling theory becomes paramount to the objective of allocating (human) resources to the firm. In passing, Spence himself has condemned the ‘flooding of the system’ with ubiquitous models that are

trained on poorly obtained data, in what Spence refers to categorically as models which are “rubbish” (Spence *et. al.*, 2024: p. 1985) in a recent interview.

Materials and Methodology

Introduction

Making use a 10-point Likert scale (where 10 = strongly agree, 1 = strongly disagree, and 5 = neutral), this thesis surveys a total of 1,034 former (in-work) job market candidates across numerous geographies using data gathered through multiple years of global content activity performed by the author. A 10-point Likert scale is justified because it is a quantitative and ordinal measure of respondent behaviours, attitudes and opinions. As stated earlier in this thesis, the purpose of the participant survey is primarily to ascertain how each respondent perceives the practice of inclusive interview accommodation and specifically measures the meaning of this to the respondent by using a value scale from 1 to 10. Here, we place emphasis on the journey of the respondent as an interview candidate. Then, by using the *IMRAD* scientific research format, the methodology of the thesis applies guidance from the popular *Econometric Analysis* (5th Edition) textbook (Greene, 2003: p.403) and prior mathematical inference published in a working paper by the author (Mayaki, 2024) to structure a 2SLS regression estimating the *nonlinear* value of the expected wage regressed upon education with endogeneity. We mimic the model from the linear case in Mayaki (2024: p.8) which outlines a ‘base case’ Pareto-Nash result excluding productivity types but demonstrating a Taylor series approximation. Firstly, we regress a vector of the ‘*expected wage*’ fitted data variable E_t , occurring across all $N = 1,034$ participants, in survey form (a response to a current salary question) at time t for all respondents given their ‘level of education’ (this is also a preliminary survey question based on: undergraduate, postgraduate, and PhD - as optional responses), which is a variable representation of the survey response concerning one’s wage utility relative to one’s current level of education at time t , and secondly, an above market wage which firms have set to attract productive cohorts of workers.

Research Philosophy

Making use of a deductive approach, the narrative review will, unlike a systematic review, focus on the

formation of an evidence-based argument using an *interpretivist* research philosophy. This is juxtaposed to an approach which builds on scientific knowledge by analysing what we might call a wide variety of contributions to a specific field, using a *positivist* methodology. Later, in the 3rd chapter of this thesis, similar techniques are applied via the use of quantitative methods (a participant survey with an operative Likert scale component and a 2SLS regression with an instrumental variable).

Research Approach

This thesis conducts secondary research by focusing on quantitative methods via a 2SLS estimation using cohort data (panel data) from the Office of National Statistics (ONS).

This disagreement in wage expectations aligns accordingly with the utility function of the worker discipline device model by Shapiro-Stiglitz, a theory employed in Mayaki (2024) describing how firm F 's cost function and offer strategy (for the *linear* case i.e. homogenous workers) minimises the likelihood of excessive worker shirking (Shapiro and Stiglitz, 1984) or to a lesser extent, avoidance of pooled workers who may carry a high probability of low productivity levels (Barmby *et al.*, 1994). Therein lies the core premise of Kamenica-Gentzkow's (KG) Bayesian persuasion framework (Che *et al.*, 2022). Table 3.1 (below) demonstrates these variables as one-half of a simultaneous equation, where the problem is *always* to balance the incentive of the workers with the motivation of the firm in a long-run equilibrium, particularly with the given likelihood of a technology shock. In Table 3.1—which illustrates the second variable in the simultaneous equation—the model assumes that the firm observes (or has *full* disclosure of) the revealed productivity of current workers. It observes productivity at time (t) and negotiates in the following stage. Firm F , then observes the profitability of their business unit and thereafter sets its wage in subsequent $t + m$ periods.

Table 3.1: 2SLS Simultaneous Equations

3.1 – What is Being Estimated in the Study?

Key Variable	Variable
Expected wage (given education)	E_t
Firm costs (wage-level data)	W_t

Source: Imbens (2014: p.328)

The study now builds on the recently published work of Miller and Doyle (2025). The nature of the newly formatted participant survey questions (See Appendices) remains consistent with Miller and Doyle (2025)'s original survey after a transformation from an open format to closed. The interrogation format was ultimately changed from open to closed to better quantify responses of workers. The ten new questions ascend from 1-10 where each item equates a rating of 1 to a strong disagreement and a rating of 10 to the highest possible level of agreement across all questions. The forte of this format is it allows structural estimates to be derived when these survey variables are embedded into an objective function (a constrained maximisation problem). The two-stage least squared (2SLS) regression model (Greene, 2003; Chen, 2019) in Equation 3.1 is defined as the dependent variable: E_t representing the *expected* wage utility of the worker at a given time, regressed on an intercept β_0 , an *observed* intertemporal level of wages variable and its estimator β_1 and X_1 , and finally a residual or error term, ε .

Instrumental Variables Using 2SLS

In Table 3.1 the 2SLS model is defined as follows:

$$(3.1) \quad \text{Expected Utility } (E_t) = \text{Intercept } (\beta_0) + \beta_{1t} \text{Wages}(X_{1t}) + \beta_{2t} \text{Dummy } (X_{2t}) + \varepsilon$$

In the 2SLS (Galeotti, *et al.*, 2025), the first stage (the instrument) identifies a dummy variable (Z) that is correlated with the message signal (the 'treatment') but is uncorrelated with the dependent variable's error term. For example, $Z = 1$ when a firm provides a list of frequently requested adjustments and $Z = 0$ when such a list is not provided, at the preliminary stage of the game. In the 2SLS second stage (the outcome), the study applies the predicted values from the first stage to estimate the impact on the dependent variable E . In terms of variables, it was assumed that one key variable emerged as a primary source in the regression function. The dependent variable E is derived from a secondary source of data. These inferences, in part, form questions which in turn are separate to the research question and are therefore secondary in nature.

Table 3.2: The Methodology

Table 3.2: Further Methodological Approaches

Handling outliers and skewness	Post-normalisation Competitiveness index
--------------------------------	--

Addressing assumption violations	If fails homoskedasticity Then robust estimation
Measurement scale adjustments	Ordinal (rank based) Interval (numeric)

The importance of the survey's applicability should not be understated, because it seeks to enquire into how diverse potential workers perceive their roles in the recruitment process *prior to* being onboarded as employees (in other words, it asks how workers interact with established incentive structures) through the proxy of possible firm strategies, actions and activities. The seminal applied instance of this research topic is the online ratings and signalling paper pioneered by Chang and Chin (2018) who test evidence to investigate the hypothesis that three things are likely to occur prior to onboarding. Firstly, a high-commitment strategy is pursued and that firms conduct all actions online via a platform. This assumes a finite degree of discretion is made on the part of the firm/sender. Secondly, the firm may initiate a signal to many receivers via a website. This signal communicates a list of reasonable adjustments (interview accommodations) which indicates the firm's high-commitment strategy. This signal initiates an effect on the employees' online ratings.

Outlining the Research Hypothesis

The authors (Chang and Chin, 2018) argue for five important hypotheses which form the starting point for this study:

- **Hypothesis 1:** Firms with more high-commitment HRM will signal more HRM information on websites.
- **Hypothesis 2:** Firms signalling more HRM on websites will see a trend of more applications.
- **Hypothesis 3:** Firms' signalling of HRM fully mediates the effect of commitment HRM on recruitment.
- **Hypothesis 4:** Firms' *actual* HRM influences employees post-experience outcomes.
- **Hypothesis 5:** Firms' signalling of HRM will not increase an employees' online ratings if the effect of commitment HRM (on recruitment) is controlled.

Regarding the hypothesis, this thesis is partly covered by *signalling* (Spence, 1973) and *labour market analysis* (Acemoglu, *et. al.*, 2008; Pissarides, 2017). My research will

aim to hypothesise concerning a gap in the signalling topic by focusing on peer effects, strategic capabilities, and firm resources, asking a question not currently addressed by the literature on signalling – how do these factors affect competitive advantage? The following hypotheses are subject to supervisory changes and will be honed and amended as and when new information in the literature is discovered, reviewed and adopted. In order, these initial hypotheses are:

- **Hypothesis 1:** The average worker views the firm's intent to accommodate as inclusive.
- **Hypothesis 2:** Firms signalling high-commitment HRM practices have happier workers.
- **Hypothesis 3:** Firms signalling high-commitment HRM strategies tend to be more inclusive.
- **Hypothesis 4:** Firms harnessing the influence of AI can achieve high-commitment *and* be optimally inclusive.

As stated earlier, the null hypothesis of no correlation holds for the thesis wage utility is correlated with surveyed skill proficiency. As there needs to be two questions so decision-makers can stay ahead of the signal. The first question is in a separating equilibrium with known payoffs, and standard assumptions (complete information and perfect monitoring), how do firms successfully navigate a many-to-one 'Spencian lottery' nonlinear equilibrium scenario, using the wage utility as an optimal signal? Secondly, but no less integral, one must ask. What is the sub-game Pareto-optimal signal, in a game of separating or pooling equilibrium (Kolotilin, 2024) that a firm should send to all workers in order to be considered inclusive? Finally, with respect to the firm, at what level of signal does a firm who signals acceptance of a health condition, become accepted by skilled workers?

Justification for Selecting Topic

Now, w.r.t. the selected subtopics, inclusion, signalling and peer effects; sufficient due diligence, care, and careful consideration were each observed when narrowing down a broader topic and selecting signalling as a topic of choice. There is also a need to resolve intertextual reference between the matching function description as found in Mayaki (2024) and its peripheral proposition on Nash bargaining, invoked in retrospect of signalling theory, albeit in a nuanced and slightly muted manner. Not only is Spence's

signalling game clearly an important canonical work to most if not all the relevant authors in the scope of the 2nd chapter of this thesis but the signalling game serves to answer a question that is very topical in current affairs and geopolitical news – How will the world of work be recalibrated when the negative impact of technological displacement effects white-collar worker utility?

REFERENCES

- Acemoglu, D., Kremer, M., and Mian, A. (2008). 'Incentives in markets, firms, and governments', *The Journal of Law, Economics, & Organization*, 24(2), 273-306.
- Anderson, J., and Markides, C. (2007). Strategic innovation at the base of the pyramid. *MIT Sloan management review*.
- Bafera, J., and Kleinert, S. (2023). Signaling Theory in Entrepreneurship Research: A Systematic Review and Research Agenda. *Entrepreneurship Theory and Practice*, 47(6), 2419-2464.
- Barmby, T., Sessions, J., and Treble, J. (1994). 'Absenteeism, Efficiency Wages and Shirking', *The Scandinavian Journal of Economics*, 96(4), pp. 561-566.
- Barney, J. (1991). 'Firm resources and sustained competitive advantage', *Journal of Management*, 17(1), 99.
- Blandin, A., Boyd, J.H., and Prescott, E.C. (2015). 'Equilibrium with Mutual Organizations in Adverse Selection Economies', *Federal Reserve Bank of Minneapolis, Research Department Working Paper* 717
- Blume, A., and Ortmann, A. (2007). 'The effects of costless pre-play communication: Experimental evidence from games with Pareto-ranked equilibria'. *Journal of Economic Theory*, 132(1), 274-290.
- Celani, A., and Singh, P. (2011), "Signaling theory and applicant attraction outcomes". *Personnel Review*, 40(2), pp. 222-238
- Chang, E., and Chin, H. (2018). Signaling or experiencing: Commitment HRM effects on recruitment and employees' online ratings. *Journal of Business Research*, 84, 175-185.
- Che, Y.K., Kim, K., and Mierendorff, K. (2022) 'Keeping the Listener Engaged: a Dynamic Model of Bayesian Persuasion', *arXiv e-print*
- Chen, W. (2019) 'Income and Effort: An Instrumental Variables Approach' *Atlantic Economic Journal*, 47, pp. 485-497
- Densley, J.A. (2012). Street Gang Recruitment: Signaling, Screening, and Selection. *Social Problems*, 59(3), 301-321
- Feld, J. and Zölitz, U. (2017) 'Understanding Peer Effects', *Journal of Labor Economics*, 35(2), 387-428
- Galdin, A., and Silbert, J. (2025). 'Making Talk Cheap: Generative AI and Labor Market Signaling'. *arXiv preprint arXiv:2511.08785*
- Galeotti, A., Goyal, S., Jackson, M. O., Vega-Redondo, F., and Yariv, L. (2009). 'Network Games', *Review of Economic Studies*, 77(1), 218-244.
- Galeotti, A., Abraham, D., and Szeidl, A. (2025) 'Peer Effects in Entrepreneurship', *LBS Working Paper*
- Gentzkow, M. and Kamenica, E. (2011). Bayesian Persuasion. *The American Economic Review*, 101(6), 2590-2615.
- Gentzkow, M. and Kamenica, E. (2016) "A Rothschild-Stiglitz Approach to Bayesian Persuasion," *The American economic review*, 106(5), pp. 597-601.
- Greene, W.H. (2003) *Econometric analysis*. 5th ed. International ed. Upper Saddle River, N.J: Prentice Hall.
- Imbens, G. W. (2014). Instrumental Variables: An Econometrician's Perspective. *Statistical Science*, 29(3), 323-358.
- Khan, M. H., Grabarski, M. K., Ali, M., and Buckmaster, S. (2023). Insights into Creating and Managing an Inclusive Neurodiverse Workplace for Positive Outcomes: A Multistage Theoretical Framework. *Group & Organization Management*, 48(5), 1339-1386.
- Kolotilin, A. (2015). Experimental design to persuade. *Games and Economic Behavior*, 90, 215-226.
- Kolotilin, A. and Li, H., and Zapechelnnyuk, A. (2024). 'On Monotone Persuasion', *Discussion Papers 2024-05, School of Economics, The University of New South Wales*.
- Kolotilin, A., Corrao, R., and Wolitzky, A. (2025). 'Persuasion and Matching: Optimal Productive Transport', *Journal of Political Economy*, 115(4), 1334-1381
- Lado, A. A., and Wilson, M. C. (1994). 'Human resource systems and sustained competitive advantage: A competency-based perspective'. *Academy of Management Review*, 19(4), 699.
- Laker, B. et. al. (2025) 'To Change Company Culture, Focus on Systems—Not Communication', *Harvard Business Review*,
- Lambotte, M. (2025), Peer Effects in Binary Outcomes: Strategic Complementarity and Taste for Conformity With Endogenous Networks. *J Appl Econ*, 40: 608-626.
- Mayaki, A.A.B. (2024) 'Pareto-Nash Reversion Strategies: Three Period Dynamic Co-operative Signalling with Sticky Efficiency Wages', *SSRN Working Paper* (Self-reference)
- Mayaki, A.A.B. (2025) 'Pareto-Nash Allocations under Incomplete Information: A Model of Stable Optima' *SSRN Working Paper* (Self-reference)

- Miller, H. and Doyle, N. (2025) 'Signalling a diversity climate: Neurodivergent experiences and perceptions during selection and assessment', *Journal of occupational and organizational psychology*, 98(3).
- Pissarides, C. A. (2017). *Equilibrium Unemployment Theory*. 2nd edn. Cambridge, MA: MIT Press.
- Porter, M. E. (2008). The five competitive forces that shape strategy. *Harvard business review*, 86(1), 78.
- Rousseau, D. M. (1985). Issues of level in organizational research: Multi-level and cross-level perspectives. *Research in Organizational Behavior*, 7, 1–37
- Scott-Phillips, T. C., Kirby, S. and Ritchie, G. R. S. (2009) 'Signalling signalhood and the emergence of communication', *Cognition*, 113(2), 226-233
- Shapiro, C., and Stiglitz, J. E. (1984). Equilibrium Unemployment as a Worker Discipline Device. *The American Economic Review*, 74(3), 433–444.
- Skardal, P. S., and Adhikari, S. (2019). Dynamics of nonlinear random walks on complex networks. *Journal of Nonlinear Science*, 29(4), 1419-1444.
- Smith, A., van Wagoner, H. P., Keplinger, K., and Celebi, C. (2025) 'Navigating AI Convergence in Human–Artificial Intelligence Teams: A Signaling Theory Approach', *Journal of Organizational Behaviour*, 0:1-31
- Spence, M. (1973) 'Job Market Signaling,' *The Quarterly journal of economics*, 87(3), pp. 355–374.
- Spence, M., Behar, A. and Jayadev, A. (2024) 'Artificial Intelligence in the Age of Uncertainty: Michael Spence in Conversation with Anurag Behar and Introduced by Arjun Jayadev,' *Information systems frontiers*, 26(6), pp. 1971–1988
- Van Damme, E., and Güth, W. (1989) 'Equilibrium selection in the Spence signalling game', *CentER Discussion Paper*, Unknown Publisher,
- Van Damme, E., and Güth, W. (1991). *Equilibrium Selection in the Spence Signaling Game*. In: Selten, R. (eds) *Game Equilibrium Models II*. Springer, Berlin, Heidelberg.
- Van Riel, C.B.M., and Fombrun, C.J. (2007). 'Essentials of Corporate Communication: Implementing Practices for Effective Reputation Management' (1st ed.). Routledge.
- Varian, H. R. (1992) 'Microeconomic Analysis: Third Edition', London & New York: Norton and Company, Inc.
- Wagener, F. (2013). 'Shallow lake economics run deep: Nonlinear aspects of an economic-ecological interest conflict'. *Computational Management Science*, 10(4), 423-450.